

HEMASREE JONNALAGADDA

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LinkedIn | Portfolio

CAREER OBJECTIVE

I'm an Embedded Systems Engineer passionate about building real-time, sensor-driven hardware solutions for applications in defense, healthtech, and industrial IoT. What sets me apart is my ability to seamlessly merge technical proficiency with a collaborative mindset. My specialization involves developing novel solutions for dynamic environments, with a particular emphasis on Firmware development.

EDUCATION

BVRIT Hyderabad College of Engineering for Women Electronics and Communication Engineering CGPA: 8.15	2020 – 2024
Deeksha Junior College Intermediate CGPA: 10	2018 – 2020
Sree Vidya High School 10th Standard CGPA: 10	2017 – 2018

TECHNICAL SKILLS

Programming & Firmware	C, C++, Embedded C, FreeRTOS, Bare-metal, BSP & HAL, DAL Design, MISRA-C, PIO, ESP-IDF, STM32CubeIDE, Pico SDK, Embedded Linux
Platforms	ESP32, STM32, RP2040, Vicharak (RP2040 + FPGA), Raspberry Pi 4, Arduino
Protocols & Peripherals	UART, SPI, I ² C, CAN, AXI-lite / MMIO, ILA, USB, GPIO, ADC, DAC, PWM, Iridium RockBLOCK 9603
Domains	Signal Processing, Telemetry, Aerospace, Healthcare, Satellite / SBD, Sensor Integration, IoT, R&D Prototyping
Tools & Simulation	MATLAB, LTspice, LabVIEW, Proteus, OpenOCD, GDB, Vivado, Git, Linux, OpenCV, GStreamer

EXPERIENCE

Catalyx Space Embedded & Ground Station Engineer	Nov 2025 – Present Gujarat, India
- Created a modular, dual-axis ground-station tracking system with embedded firmware for ACU autonomy and RF coordination. - Implemented degree-to-pulse conversion, RPM-based velocity control, and fractional pulse accumulation for high-resolution tracking. - Developed real-time DAQ and telemetry systems for rocket propulsion testing with high-speed data acquisition. - Designed reliable frame parsing and USB CDC stream reconstruction for stable data logging under fragmented conditions. - Built low-latency multi-camera imaging pipelines using C++, OpenCV, and GStreamer for real-time monitoring. - Implemented asynchronous multi-camera capture with thread-safe processing and automated recovery for high performance.	
Geminex Technologies Freelance Firmware Engineer	Jun 2025 – Aug 2025 Gujarat, India
- Developed embedded firmware for a digital stethoscope on STM32. - Integrated ECG (AD8232), SpO₂ + Heart Rate (MAX30102), Temperature (MAX30205), and Digital Mic (INMP441) sensors for real-time physiological signal monitoring. - Implemented ADC data acquisition and real-time ECG and microphone waveform plotting.	
IIT Indore (Drishti CPS Fellowship) Embedded R&D Engineer	Oct 2023 – Dec 2024 Indore, India
- Designed and built an embedded wearable using ESP32-S3 integrated with BNO055 IMU and Kinect v2 for gait data acquisition. - Programmed the ESP32-S3 for I²C communication and Wi-Fi transmission of motion data. - Worked on sensor interfacing, deep learning model testing, and data collection for gait-analysis experiments.	

PROJECTS

FPGA-Accelerated Real-Time Control System | Vicharak (RP2040 + FPGA)

- Designed a layered device driver architecture with a DAL (Driver Abstraction Layer) modeled after Linux driver patterns — unified open/read/write/ioctl ops table across all hardware.
- Wrote custom drivers from scratch: FPGA register-map driver (memory-mapped), PIO high-speed bridge driver, interrupt-driven UART driver with FreeRTOS queue backend.
- Designed custom FPGA IP blocks (PWM engine, signal capture, register map) and a firmware-side ILA readback driver over SPI for in-circuit signal capture.
- Debugged RP2040 + FPGA simultaneously using OpenOCD/GDB with FreeRTOS task-aware breakpoints and FPGA-side logic analyzer.
- Implemented FreeRTOS trace hooks (vTaskList, runtime stats) with a CLI task for live task monitoring over UART.

Skills: FreeRTOS, Driver Abstraction Layer, Custom Device Drivers, PIO, FPGA IP Design, OpenOCD, JTAG, GDB, ILA, Memory-mapped I/O, RP2040, Vicharak

Embedded Fault-Tolerant Attitude Monitoring System

- Built a fault-tolerant attitude monitoring system using STM32F303 and BNO055 IMU with real-time orientation tracking and automated anomaly recovery.
- Engineered robust error detection algorithms for sensor freeze, signal spikes, and disconnection events with system state visualization using onboard LEDs.
- Logged telemetry data over UART and simulated aerospace-grade reliability through watchdog timers and redundant attitude cross-validation.

Skills: GY-906, Sensor Interfacing, Microcontroller Programming, IMU, Serial Communication, STM32F303, BNO055, ESP-IDF

Anomaly Detection for Hot Axle Monitoring

- Developed a pipeline for real-time detection of Hot Axle condition in railway systems using sensor time-series data (temperature, humidity, vibration axes).
- Applied a Kalman Filter to extract true system state from noisy sensor data, enabling reliable fusion and filtering of vibration and temperature data.
- Implemented unsupervised learning (DBSCAN, Hierarchical Clustering) to classify system states into Normal, Warning, and Critical without labeled data.

Skills: Esp32c3, Kalman filter, Arduino IDE, MATLAB, DBSCAN, Hierarchical Clustering

Gait Revive - Beyond the walk

- Developed an end-to-end intelligent wearable and software platform for real-time gait assessment and therapy.
- Engineered a custom wearable device embedded with motion sensors to accurately capture positional data movements in real time.
- Integrated Kinect-based motion capturing with wearable sensor data to enhance spatial resolution and improve detection accuracy of gait abnormalities.
- Developed a virtual therapy module delivering tailored rehabilitation programs based on dynamic movement patterns and improvement tracking.

Skills: Esp32c3, I²C Communication, BNO055, C++, Xbox - one, Kalman filter, Arduino IDE, MATLAB, Firebase, Deep Learning, Web Dev

CERTIFICATIONS / ACHIEVEMENTS

- **Approved idea under MSME Idea Hackathon 4.0** — “Diagnosis of Abnormal Walking Patterns using Deep Learning” for **Gait Revive** selected under the MSME Innovative Scheme (Incubation Component), Government of India, with a **sanctioned project cost of 15.00 lakhs** for wearable gait diagnosis and rehabilitation.
- **Secured a 1 lakh UG fellowship grant** for the “Gait Revive” project under Drishti CPS, IIT Indore, recognized in the **BUILD program** by IIT Hyderabad’s iTIC Incubator, winning 1 Lakh and crucial support. Achieved **second place** and a 1.5 Lakh cash prize in the **Take Off competition** by UPES and Runway Incubator for “Gait Revive.” Presented the project at **TSIC Summit 3.0** and introduced it at the International Conference on IPR and SME, sponsored by the Ministry of MSME.
- Won the **FIRST PRIZE** for the idea “**e-Setch Bin**” in Social Innovation Hackathon organized by Telangana State Innovation Cell (TSIC) in T-Innovation Mahotsavam.
- A **patent published** for the projects “**Gait Revive**” and “**Sight Scribe**” marking a significant step towards protecting and securing intellectual property rights for this innovative initiative.
- **Presented a research paper** titled “Smart Vest for Women Undergoing Menopause” organized by Springer.
- Attended a **2 day workshop** organised by Vertulonix on IoT devices.
- Participated in a **mentorship program** organized by Qualcomm.
- **Certified in MATLAB Onramp:** Proficient in MATLAB programming, data analysis, and algorithm development.

LEADERSHIP EXPERIENCE

- As the primary point of contact, I fulfilled a crucial function as the coordinator between our college and IIT Madras within the **Build Club**. During my tenure between December 2022 and March 2023, I oversaw the coordination of various projects.
- I successfully organized and coordinated the **HACK2EMPOWER** and **PROCREATE** event held at our college as a component of the National Tech Fest, **MEDHANVESH**.